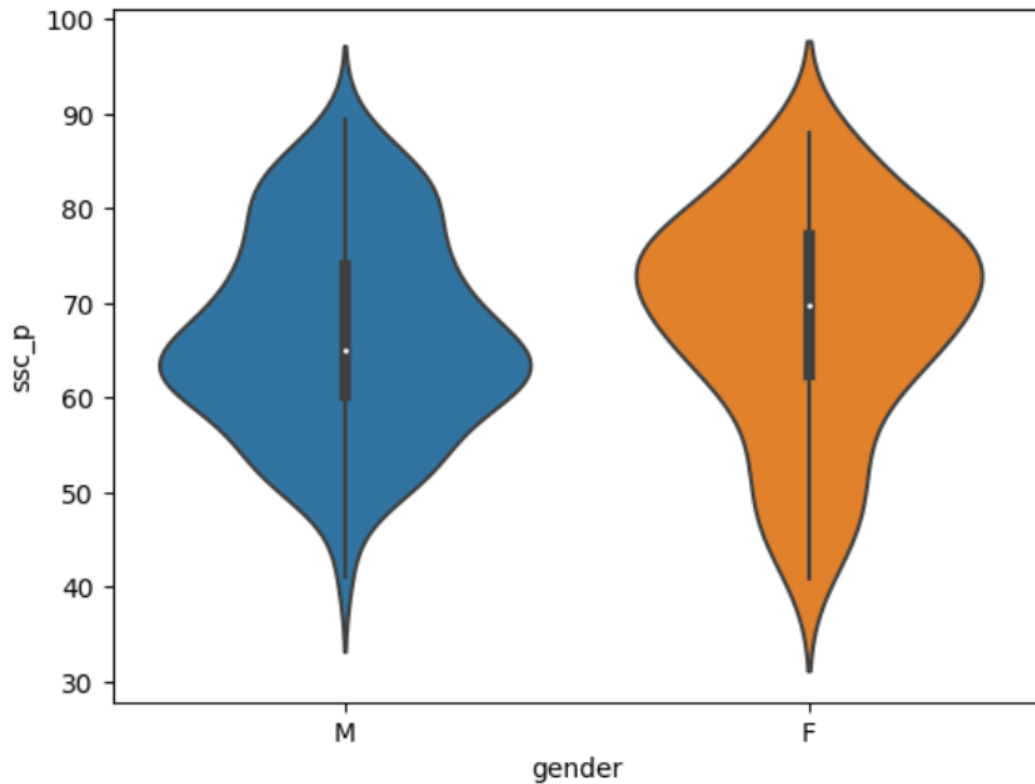


Visualization of 10th Marks (SSC_P Marks) Using Violin Plot

```
sb.violinplot(x = "gender", y = "ssc_p", data = dataset)  
plt.show()
```



Assignment Perspective – Questions & Answers

1. What does the X-axis represent?

The X-axis represents the gender category of students:

- M = Male
- F = Female

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2. What does the Y-axis represent?

The Y-axis represents SSC percentage (ssc_p), which indicates 10th standard marks.

3. What is the purpose of a violin plot?

A violin plot is used to visualize:

- Distribution of data
- Density of values
- Spread of observations
- Median and variability

4. Which gender has a higher average SSC percentage?

Female students appear to have a slightly higher average SSC percentage compared to male students.

5. Which gender shows more variation in marks?

Male students show more variation because the violin shape is wider across multiple ranges.

6. Around which mark range are most students concentrated?

Most students are concentrated between:

- Male: 60–70
- Female: 65–75

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7. What does the width of the violin indicate?

The width indicates the density or concentration of data points.

- Wider area → more students in that mark range
- Narrow area → fewer students

8. What does the white dot inside the violin represent?

The white dot represents the median (middle value) of the data.

9. Which gender has more consistent marks?

Female students appear to have more consistent marks because their distribution is more concentrated.

10. What can be concluded from this graph?

The graph shows that female students generally scored slightly higher SSC percentages, while male students showed greater variability in marks.

11. Why do we use a violin plot?"

Violin plots are used to understand the distribution and density of numerical data across different categories.

Conclusion:

The violin plot compares SSC percentages between male and female students. Female students show slightly higher and more consistent marks, while male students show greater variation in SSC percentages.